ZAIN TAMER ZAIN ELABDIN

Full CV

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Summary

Computer Engineering student at AAST (CGPA 3.99/4.0) specializing in machine learning, computer vision, and MLOps, with hands-on experience building production AI systems deployed on AWS and Azure. The algorithmic rigor behind that work — shaped by years of competitive programming — carries across every model architecture and system design decision. Equally driven by working alongside others — whether teaching, mentoring, or building as part of a team — as by the technical craft itself. Currently directing research interests toward large language models and NLP, with a focus on how these systems can be grounded, adapted, and deployed in production.

Education

Arab Academy for Science, Technology & Maritime Transport University — Alexandria, Egypt
Bachelor of Science in Computer Engineering — Sep 2022 – Feb 2027
CGPA: 3.99/4.0
Relevant Coursework: Discrete Mathematics, Probability & Statistical Analysis, Artificial Intelligence, Data Analytics and Optimization, Database Systems, Computing Algorithms, Object-Oriented Programming, Data Structure & Algorithms, Distributed and Parallel Systems, Embedded Systems Design, Cyber Security, Intro to Intelligent Human Computer Interaction.

Experience

Machine Learning Engineer Intern – Digital Egypt Pioneers Initiative — Jun 2025 – Dec 2025

- Completed a structured **180-hour** applied ML programme covering data engineering, computer vision, NLP, and cloud deployment, with hands-on delivery across all domains.
- Built and deployed machine learning pipelines for computer vision and predictive analytics tasks, managing the full workflow from data preprocessing and feature engineering to model serving on **Azure Machine Learning** and **Azure App Service**.
- Developed NLP pipelines for text classification and sentiment analysis, applying Transformer-based models to real-world text datasets and managing artifacts on **Azure Blob Storage**.

Projects

Premier League Predictor – End-to-End MLOps System

- Engineered a full ML pipeline using **ZenML** to fetch historical match data, train a **XGBoost** classifier, predict upcoming fixture outcomes, and project the final league table, with a **Streamlit** dashboard for results.
- Integrated **MLflow** for experiment tracking and model registry, automatically promoting the highest-accuracy model to a *champion* alias and storing artifacts on **AWS S3**.
- Stored and retrieved processed datasets from a Supabase **PostgreSQL** cloud database, enabling decoupled pipeline execution.
- Containerised the system with **Docker**, deployed to **AWS EC2** via **AWS ECR**, and scheduled weekly automated evaluation via cron — triggering a full retraining pipeline when live prediction accuracy dropped below threshold.

FPL Vision – AI-Powered Fantasy Premier League Assistant

- Built a full-stack web assistant for Fantasy Premier League managers, generating player recommendations and expected points projections using a trained **XGBoost** model.

- Aggregated and processed data from multiple sources including the **FPL REST API** across **700+ players and 20 teams**, engineering features from historical performance, fixtures, and player form to drive model predictions.
- Deployed the **Streamlit** application on **Azure App Service**, with model artifacts and processed datasets managed on **Azure Blob Storage**, surfacing live dashboards for player and team statistics; used **GitHub Actions** for CI/CD to automate deployment on data change and code updates.

Facial Recognition System

- Built a real-time facial recognition pipeline using a pre-trained **FaceNet** backbone with a fine-tuned transfer learning classification head trained on the **LFW dataset** (303 identities, 640 embeddings) achieving **98% accuracy**.
- Implemented multi-face detection via **MediaPipe**, generating 128-dimensional L2-normalized embeddings with cosine similarity matching, and a user registration flow supporting new identity enrollment.
- Containerized and deployed the **Streamlit** web application using **Docker** on **Hugging Face Spaces**.

WHO COVID-19 Global Daily Data Analysis

- Performed an in-depth exploratory data analysis (EDA) on WHO's global COVID-19 dataset comprising greater than **250000** daily records across **200+** countries/regions.
- Focused on **data cleaning, feature engineering, and statistical visualization** to uncover insights about global case trends.

Chat Room

- Engineered a **LAN-based real-time** chatroom application, supporting text, voice, and video communication using **Python** and **socket programming**.
- Designed the application to handle up to **10 concurrent users** with seamless media switching.
- Gained hands-on experience with **socket programming, client–server architecture, multithreading, and low-latency network communication.**

Autonomous Point-to-Point Smart Car

- Developed an autonomous ground vehicle capable of navigating to user-specified coordinates in a 2D plane using onboard **IMU**-based localization and motion control on **Arduino**.
- Integrated **MQTT** communication and a **voice recognition** service to enable wireless and voice-based command of target positions and vehicle actions.

Research

Research Project: An Experimental Analysis of Data Augmentation and Hyperparameter Tuning for Image Classification — Oct 2025 – Dec 2025

- Conducted an expiremental analysis on various sets of augmentation techniques on different deep learning architectures with proposing new augmentation methods and studying their contribution to the model performance.

Research Project: Comparison of Quicksort and BFPRT Algorithms for the K-th Element Selection Problem — Feb 2025

- Investigated different solutions for the **selection problem for the k-th element.**
- Done a research on the **performance** of those different solutions with respect to **the size of the problem.**

Skills

Programming Languages: Python, C, C++, C#, Java, SQL, Bash
Machine Learning & Data Science: Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, Plotly, TensorFlow, Keras, PyTorch, Transformers, OpenCV
Cloud & DevOps: Microsoft Azure, Amazon Web Services, Git, GitHub, PostgreSQL, huggingface spaces, Linux
Deployment & MLOps: ZenML, MLflow, Docker, CI/CD, GitHub Actions, Kubernetes
Additional Tools: Hadoop, Arduino, MQTT, LaTeX

Algorithms & Problem-Solving: Data structures, graph algorithms, dynamic programming, time and memory optimization
Soft Skills: Leadership, Communication, Teamwork, Problem-Solving, Flexibility, Resilience, Multitasking
Languages: Arabic (Native), English (Advanced), German (Beginner)

Certificates

Codeforces Specialist (max rating: **1448**) Nov 2025
ARTIFICIAL INTELLIGENCE ENGINEER 1 (Coursera – IBM) Oct 2025
Sprints x Microsoft Summer Camp – AI and Machine Learning Oct 2025
ECPC (Egyptian Collegiate Programming Contest) Qualifications Jul 2025
Zindi Financial Inclusion in Africa Competition Jun 2025

- Ranked **15th** on the leaderboard out of over **2000** participants.

IBM Software Developer Roadmap (Coursera – IBM) Nov 2024
IEEEXtreme Programming Competition Oct 2024

- Ranked in top **2%** in the Egyptian region and in top **20%** worldwide from over 8000 teams.

Volunteering

AWS Student Builder Group at AAST - Team member Mar 2026 - Present

- Participated in organizing several cloud practicioner training sessions.
- Lead a workshop on different AWS services for 20+ students.

AAST Competitive Programming Club – Coach Sep 2025 – Present

- Led **15+** tutoring sessions explaining competitive programming fundamentals and algorithms.
- Monitored and coached a group of **30+** students.
- Contributed in preparing materials and contests for trainees.